From Keyword Hits to Relevant Passages

Finding the European Idea in the Noisy German-Language Press of Interwar Romania

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Abstract

Historians of the interwar press search digitised archives by keyword. They then read every returned page to find the articles that discuss their topic. On Fraktur OCR this step is slow, and keywords miss words the OCR misread and articles without the keyword. We ask how far ranking can shorten it. Our case is the idea of European unity in the German-language press of interwar Romania, 1926 and 1929. A historian marked every relevant article on 20 keyword-selected and 10 random pages, blind to all systems. We compare five methods that rank the passages of these pages, from an archive-style keyword to dense retrieval, against reading order. Our main measure is reading effort: how many passages a historian reads to find 80% of relevant articles. An open-weight language model (Qwen3-30B-A3B) then states the stance of each relevant article with a reason and a quote, which we check against the text. On the keyword-selected pages, the archive keyword needs the least reading: 35 passages for 80% of the 15 relevant articles, against 444 unranked. Dense retrieval gives the cleanest top ten, but none of the three articles without the keyword reaches its top 20. On the random pages the keyword finds nothing, while the other methods rank the one relevant article in their top four. The model agrees with the historian on 7 of 15 stances; it takes reported speeches and French–German reconciliation for support, and 4 of 16 quotes are not verbatim. We release the guideline, page lists and code.

CCS Concepts

• **Information systems** → **Evaluation of retrieval results**; • **Applied computing** → *Arts and humanities.*

Keywords

historical newspapers, OCR noise, passage retrieval, stance detection, large language models, digital humanities

1 Introduction

Digitised newspaper archives have changed how historians work with the press: papers that were hard to reach and slow to search through are now a keyword search away [1]. A typical session on a commercial archive such as Arcanum runs as follows. Historians type a term such as "Paneuropa", download the pages that match, and read each page to find the article that really discusses the topic. They then work with the article, through notes, excerpts or a transcription depending on their question, and interpret what it says. The keyword search is fast, but the reading afterwards is not.

Two problems make this reading costly. First, the text behind the search is OCR output, and on old print it is noisy. Historians know that OCR errors may bias their results, but they cannot say by how much [29]. In our Fraktur pages, "Paneuropa" appears as *Panzuropa* or *baneuropa*, and "Völkerbund" as *bölferbund*. Figure 1 shows the scale: a quarter of the occurrences of "Paneuropa" in our pages are so garbled that no keyword method finds them.

Second, a keyword finds a word, not a topic. Many hits name Europe only in passing, while an article can argue for European cooperation without the word "Europa" at all. We call an article that discusses the idea of European unity an IDEA article, and one that only names Europe or Paneuropa a MENTION (Section 5).

Our case is the idea of European unity between the wars. Richard Coudenhove-Kalergi's Paneuropa movement called for a "United States of Europe" and held its first congress in Vienna in October 1926 [9, 11]. In September 1929, the French foreign minister Aristide Briand took the idea to the League of Nations and proposed a federal bond among the European peoples [9]. Interwar Romania had a Romanian-language press and a large minority press, above all in Hungarian and German. We study German-language papers from Transylvania and the Banat, because the Paneuropa programme called for the protection of national minorities [9, 11] and because our historian reads them natively. Our results speak for these three

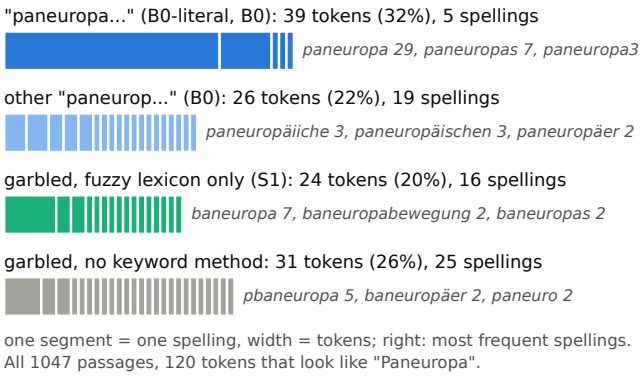


Figure 1: How "Paneuropa" comes out of the OCR. A token counts as a form of the word if its first eight letters are close to *paneurop* (normalised Levenshtein similarity of at least 0.75; all 65 spellings checked by eye). In all 1,047 passages we built (including 151 from journal extracts not used in this study), 120 tokens qualify. Only 32% start with "paneuropa", which is all a literal keyword finds; the stem *paneurop (B0) also finds the second row, 54% in all. The OCR-tolerant word list (S1) adds the third row; 26% are so garbled that no keyword method matches them. One segment per spelling, width = tokens.**

titles, not for the German minority press as a whole, let alone for opinion in Romania.

We do not try to classify the whole archive. Relevant articles are rare, and we have no labels to train on. Instead we treat the step after the keyword search as a *funnel*. A ranking puts the passages most likely to discuss the idea first, and historians read from the top. Only the articles they keep go to a language model, which states each article's stance with a reason and a quote they can check. The human stays where the judgement lies.

We ask two research questions.

RQ1 *On pages returned by an archive keyword search, and on pages nobody pre-selected, how well do five ranking methods lead a historian to the articles that discuss European unity?*

RQ2 *When an open-weight language model reads such an article, how often does its stance agree with the historian's, and is its quoted evidence really in the text?*

Our contributions are four. (1) A small gold standard: every relevant article on 30 newspaper pages, marked by a historian blind to all systems, with a public guideline. (2) A comparison of five retrieval methods on noisy Fraktur OCR, measured as the reading effort a historian saves, including articles that never use the keyword. (3) A stance reading with evidence, checked automatically against the source text. (4) A reflection on our own process: we planned a larger four-stage system, built only the part we could test, and say why. In short, on pages that an archive keyword search returned, any ranking cuts the reading by a factor of seven or more, and the archive's own keyword is hard to beat. On random pages that nobody selected, the keyword finds nothing, while the word lists and dense retrieval put the one relevant article near the top. Articles that never use the keyword stay hard for every method. The language model agrees with the historian on 7 of 15 stances: it finds every PRO article, but it reads 7 of 9 NEUTRAL articles as PRO, mostly speech reports and French–German reconciliation.

2 Related Work

We group prior work by the six aspects our study touches: the history we look for, historians' use of digitised newspapers, the infrastructures built for them, OCR noise in retrieval, the retrieval methods we compare, and stance detection with language models.

2.1 The European idea between the wars

The idea we look for has a clear interwar history. Richard Coudenhove-Kalergi founded the Paneuropa movement and saw Europe as a community of culture and values rather than a territory with fixed borders [9]. His model for a "United States of Europe" owed much to the multinational Habsburg Empire in which he grew up [9]. The Paneuropa Union was founded in Vienna in 1925 and held its first congress there in October 1926, opened by the former Austrian chancellor Ignaz Seipel [11]. In August 1929, the Hague Conference adopted the Young Plan for German reparations [26]. A few weeks later, Aristide Briand proposed a federal bond among the peoples of Europe before the League of Nations Assembly [9, 11]. Federal plans for the region were older: in 1906, the Romanian Aurel C. Popovici had proposed turning the Habsburg Monarchy into a "United States of Greater Austria" [21]. Most of our keyword pages come from the two moments of 1926 and 1929 (Section 5). *What this leaves open for us:* this history tells a historian what to look for, but not where the debate appears in thousands of noisy newspaper pages. That search is the step our study supports.

2.2 Historians and digitised newspapers

Digitisation removed two old obstacles of press research, access and finding relevant material, but digital searching has methodological pitfalls of its own [1]. OCR errors are one: pages with more recognition errors are harder to find by search [28] (Section 2.4). Interviews with historians found four kinds of information seeking in archives, among them orienting oneself in a collection and identifying relevant primary sources [8]. Traub et al. interviewed historians about digital archives [29]. They were aware that OCR errors may bias their results, but could not quantify that bias or say what they would need to estimate it. Pfanzelter et al. review the interfaces of newspaper portals [24]. They ask which tools help which user groups without adding too much complexity. Two further lines ask how scholars can trust a tool. Koolen et al. argue that tools which go beyond keyword search transform the material and call for a *digital tool criticism* [19]. El-Hajj et al. argue for explainable AI in the humanities, so that historians can read and reproduce what a model does [12]. *What this leaves open for us:* these works describe the reading burden and the need for tools a historian can question. They do not measure how much a better ranking reduces the reading, nor test which kind of explanation a ranking should show.

2.3 Infrastructures for historical newspapers

Two large projects built tools for historical newspapers. Impresso published corpora, benchmarks, annotations and language models in French, German and Luxembourgish, covering about 200 years [10]. NewsEye built tools for newspapers in German, Finnish, Swedish and French from 1850 to 1950 [7]. Its tools range from text recognition and article segmentation to named entities, stance, topic models and a personal research assistant. Humanities scholars in the project ran case studies, among them on nationalism. Its dataset for named entities, entity linking and stance is the closest resource to ours [13]. *What this leaves open for us:* the stance in that dataset is taken toward named entities, not toward an idea such as European unity. The corpora also come from other regions; to our knowledge neither covers the press of interwar Romania.

2.4 OCR noise and retrieval

OCR errors reduce what a search can find. In a historic newspaper collection of a digital library, the character error rate of a document correlated strongly with its retrievability, and manual correction made retrieval less biased [28]. For the Gallica library, Chiron et al. estimated the risk that a user's query terms miss documents because of OCR errors [2]. Van Strien et al. found a consistent effect of OCR quality on several downstream tasks, retrieval among them [30]. Hill and Hengchen compared an eighteenth-century OCR corpus with its hand-keyed version and measured how the errors change topic models, collocations and vector-space results [14]. De Oliveira et al. compared OCR and correction methods on a Portuguese test collection [6]. The OCR method changed mean average precision

by up to 5 points, while correction changed little on average. Correcting OCR with language models is no free fix either: it reduced errors for English but did not reach a useful level for Finnish [17]. Two ways around the noise follow from this work: correct the text first, or search in a way that tolerates the errors. We take the second, keep the OCR noise, and measure retrieval against it. *What this leaves open for us:* these studies work with whole documents in large collections, in English, French, Portuguese and other languages. They measure what a search finds, not how much a reader must read. We found no study that ranks the passages *within* pages a keyword search has already returned, on German Fraktur, or that compares an OCR-tolerant word list with dense retrieval there.

2.5 Retrieval methods and their evaluation

Our methods come from three families, and each has a reason to be tried here. *Lexical ranking* with BM25 scores a passage by how often it contains the query terms, weighted by how rare they are [25]. Across 18 retrieval tasks, BM25 stayed a dependable baseline, while dense models generalised worse to new domains [27]. We have no training data from our domain, so BM25 is the natural lexical baseline. *Approximate matching* finds words that differ from the query by a few letters, usually measured by edit distance [22]. It is the classic answer to spelling variation, and OCR errors are spelling variation on a large scale. *Dense retrieval* compares passages and queries as vectors, so it can match meaning without shared words. Dense passage retrieval beat BM25 clearly in open-domain question answering [18], and multilingual E5 extends such models to many languages [32]. But dense models also lose much of their effectiveness when the query contains typos [34]. Reciprocal rank fusion combines rankings by their ranks alone; in TREC experiments it beat each single system [5]. For evaluation, nDCG credits graded relevance and discounts late hits [15]. Closer to a historian's view is Cooper's expected search length: how many items a user must examine to find the relevant ones she needs [4]. Relevance judgements vary between assessors, yet system comparisons were stable despite this [31]. *What this leaves open for us:* the evidence for each family comes from modern, mostly English text, and the typo studies put the errors in the query, not in the documents. Whether edit-distance matching or dense vectors cope better with OCR noise in the documents, and on German Fraktur, is open. These measures also count documents, while the historian's unit is the article, which the OCR layer does not mark. We therefore let methods rank passages but count articles, and we report reading effort in the spirit of expected search length.

2.6 Stance detection and language models

Stance detection classifies a text's position toward a target, usually as favour, against or neither [20]. The task grew with user-generated content on social networks, news sites and discussion boards [20]. Large language models now allow zero- and few-shot stance detection. A recent survey names implicit stance as an open challenge, and explainable stance reasoning and low-resource adaptation as future directions [23]. Language models can also produce fluent text that is not supported by their input [16]. A quoted piece of evidence is thus only useful if it is checked. *What this leaves open for us:* we found no study of stance toward an abstract idea in noisy historical text. We test this with one open-weight model from the Qwen3 family [33], zero-shot, and we check each quote against the article. We chose an open-weight model so that the run can be repeated with the same weights. It could also run at an academic computing centre instead of a commercial service (Section 4).

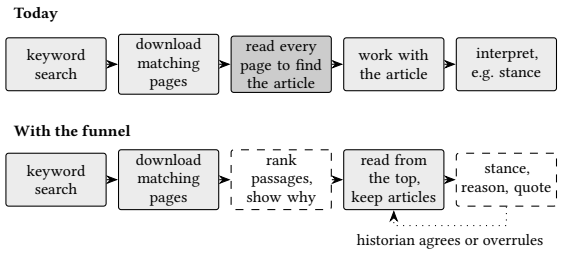


Figure 2: The historian's workflow today and with the funnel. Grey: done by the historian (dark grey: the costly step). Dashed: done by the system (RQ1 ranks the passages, RQ2 reads the stance).

3 Stakeholder Needs and Approach

Today's workflow. Figure 2 shows how our historian works today. The archive search returns pages, not articles, and a page of the *Siebenbürgisch-Deutsches Tageblatt* holds many articles in several columns. So the costly step is reading every returned page to find the article that really discusses the topic. What happens next depends on the researcher and on what she wants to do with the article: she may take notes, excerpt passages or transcribe it in full. Our tool leaves the search and this later work as they are and changes only the costly step: it ranks the passages of the returned pages, and the historian reads from the top.

Requirements. Table 1 lists what historians asked of such a tool when we met researchers working on these topics at a conference in Budapest in summer 2026. It also gives our design for each need and how far we built it. Two needs shaped the design most. The historians want to see why a passage surfaced, and they want reasons rather than verdicts. Every method therefore returns an explanation for each passage, such as the matched OCR variant *bölferbund∼völkerbund* or the nearest concept description. The stance comes with a one-sentence reason and a quote the historian can find on the page.

Why a funnel and not a classifier. A classifier over the whole archive would need many labelled examples, and passages on the European idea are rare. We had no labels, and a verdict on every page would be hard for a historian to check. A funnel instead ranks the pages the historian already has, so she reads less but still reads. Only the small set she keeps goes to the language model, and each stance comes with its evidence. This keeps a human at the point where judgement matters, and it needs labels only to evaluate, not to train.

Scope. We build one thin slice of the planned system: the ranking step and the stance with evidence. We leave out OCR correction,

Table 1: Historians' requirements, our design, and its status in this paper.

Need	Our design	Status
Find the article that discusses the topic, not every hit	rank passages, count articles (RQ1)	built
Find words the OCR misread	OCR-tolerant word list (S1)	built
Find articles that never use the keyword	dense retrieval by meaning (S2)	built
See why a passage surfaced	matched term or nearest concept per passage	built
Reasons, not verdicts	stance with reason and quote; quote check (RQ2)	built
Trace every finding to its page	passage ids point to page and position	partly
Stance is the text's own voice; "pro Europe but against X"	guideline rule in the prompt; free `against:X` tag	partly
Collapse reprints of the same item	`reprint` tag only	future
Show OCR quality next to each result	not built	future
Rise of the debate around Briand's plan, 1929 to 1932	not possible: our pages cover a few months	future

topic clustering, the explorer interface, reprint detection, the timeline, and the comparison across the three language communities. Section 7 explains this choice.

4 System

The system has two parts, one per research question. The code for each step is in the repository[1].

From PDF to passages. Each PDF carries a text layer from Arcanum's OCR, stored as text blocks of a few lines each. We read these blocks and only normalise historical glyphs (long *s*, ligatures, quotation marks) and join words split at a line end. We do not correct OCR errors, so every method faces the same noise the historian's search faces. The OCR does not mark where an article begins or ends, so we cut the page into passages that follow its columns. We link each block to the block directly below it in the same column. Along this chain we merge blocks until a passage has at least 40 words. We start a new passage before one would pass 250 words, and a short remainder joins the passage above. A passage is thus a piece of one column, often one or two paragraphs, and it points back to its page and position. Over the two page sets we score this gives 896 passages, with a median of 52 words.

RQ1: ranking methods. Table 2 lists the methods. R0 is no method at all: the pages read in order, top to bottom, as a historian reads them today. B0 imitates the archive search: it counts the words that start with *paneurop*, so inflected forms such as *paneuropäischen* count. B0-literal counts only *paneuropa**; we added it after the first results to check whether the stem flatters B0, so it is not preregistered. B1 and S1 use the same word list of 13 terms, such as *paneuropa*, *coudenhove*, *briand*, *völkerbund* and *zollunion*. We wrote the list ourselves from the names and terms in our design dossier; Section 7 discusses what this means. Terms of five or more letters also match as prefixes, so German endings count. Both score with BM25 at its usual settings ($k_1 = 1.5$, $b = 0.75$), and the word weights (IDF) come from the passages of the page set being ranked. S1 adds tolerance for OCR errors with edit distance [22]. A word matches a term if its first letters, as many as the term has, reach a normalised Levenshtein similarity of at least 0.8. This allows at most about one wrong letter in five. For example, *baneuropa* matches *paneuropa*, with one wrong letter in nine (0.89). We chose 0.8 before any result as a bound that allows one or two OCR errors in a long term but not a different word; we did not tune it. S2 ranks by meaning. It embeds each passage and three short German descriptions of the idea, which we wrote ourselves, with `multilingual-e5-base` [32]. We chose this model because it is a multilingual retrieval model with open weights, and its base size embedded all passages on a laptop CPU in about 20 minutes. A passage scores its cosine similarity to the closest description. S3 fuses S1 and S2 with reciprocal rank fusion [5]: each method adds $1/(60 + r)$ for a passage at rank r, with the constant 60 that Cormack et al. found near-optimal. Fusion by ranks needs no calibration between BM25 scores and cosine similarities, and it gives both methods the same weight. A method adds to the fused score only for passages it actually retrieved, so the unmatched tail of S1 adds nothing. In every method, passages with equal scores keep their reading order. This is what the historian would do with them anyway, and it means that no method gains or loses from an arbitrary order. The word list, the descriptions and all settings were committed to the repository before any label came back, so we could not tune them to the answers.

Table 2: The ranking methods (RQ1). Every method also returns, for each passage, why it surfaced. R0 is the reference without a tool. †Added after the first results; not preregistered.

Id	Ranks passages by	Shown to the historian
R0	reading order (no ranking)	–
B0	hits of *paneurop**	matched words
B0-lit.†	hits of *paneuropa**	matched words
B1	BM25 [25], word list, exact spelling	matched words
S1	BM25, word list, OCR-tolerant	matched words and variants, e.g. *bölferbund*∼*völkerbund*
S2	similarity to three concept descriptions	the closest description
S3	rank fusion of S1 and S2	both explanations

Why these methods. Each step isolates one effect. B1 against B0 shows what a wider word list with BM25 weighting adds. S1 against B1 shows what OCR tolerance adds, since the word list is the same. S2 tests whether meaning finds what spelling misses, above all articles without the word "Europa". S3 tests whether the two kinds of evidence add up. Figure 3 shows what each method returns for one article: its rank and the reason it surfaced.

[1] Code in `src/popline/`: passages in `pdftext.py`, ranking methods in `retrieve.py`, measures in `metrics.py` and `rq1.py`, stance and quote check in `stance.py`. Queries and settings in `configs/`.

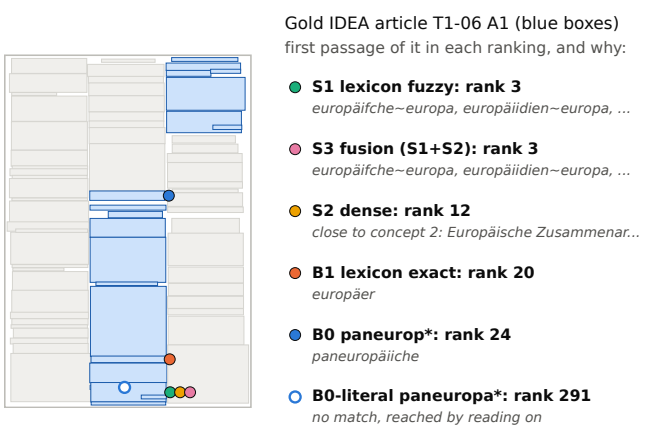


Figure 3: What the historian sees for one page (T1-06, 1926), drawn as passage boxes, not as the scanned page. Blue: the gold IDEA article on this page, an article the historian marked as discussing European unity (here a report of the Paneuropa congress). Each dot sits on the passage of this article that a method ranks highest. Its label gives that passage's rank in the method's ranking of all 608 tier-1 passages (1 = read first) and the explanation shown to the historian, e.g. *europäifche~europa*: an OCR form matched to a word-list term.

RQ2: stance with evidence. For each of the 16 articles the historian marked as IDEA (15 on tier 1, one on tier 3), an open-weight model reads the whole article. We cut its text from the PDF text layer inside her rectangles, column by column, and join its continuation where she marked one. The model sees only this text, never her label. We use `qwen3-30b-a3b-instruct-2507`, an updated instruction-tuned release of the Qwen3-30B-A3B model [33], at temperature 0, zero-shot. Its weights are open under the Apache 2.0 licence [33], so the run can be repeated with the same model, and it is offered by an academic computing centre that does not keep the text. The prompt (Appendix A) quotes the guideline's stance rule word for word: the stance is the text's own voice, so a neutral report of a pro-European speech is NEUTRAL. The model answers PRO, CONTRA or NEUTRAL, with a one-sentence reason and a quote of at most 40 words copied from the article. The model ran on SAIA (chat-ai.academiccloud.de), the AI service of GWDG, the academic computing centre in Göttingen, which hosts open-weight models on its own hardware. GWDG states that prompts and replies to these models are never stored and are not used for training[2]. This matters because the pages are subscription material; we have not audited the claim.

5 Data and Evaluation Setup

Data. Table 3 lists our page sets. All pages come from Arcanum's newspaper database (Arcanum Digitheca, https://adt.arcanum.com), downloaded during a subscription in the second half of June 2026. Tier 1 holds the extracts an Arcanum search for "paneuropa" returned; an extract is a PDF of one or more pages of an issue, so some pages carry no hit. They are 12 extracts of the *Siebenbürgisch-Deutsches Tageblatt* (Hermannstadt/Sibiu, October 1926), one of the *Temesvarer Zeitung* (Temeswar/Timișoara, January 1926) and one page of the *Banater Deutsche Zeitung* (Temeswar/Timișoara, August 1929), 20 pages in all. Tier 3 is our control: 10 pages drawn at random, with a fixed seed, from a run of 50 consecutive pages of the *Banater Deutsche Zeitung* (August 1929) that we downloaded without a keyword search. All three newspapers are set in Fraktur.

Table 3: Page sets. Passages from the ingest run; articles from the gold standard.

Tier	Selected by	Pages	Passages	IDEA	MENTION
1	keyword "paneuropa"	20	608	15	6
3	random (control)	10	288	1	0

Gold standard. Our historian, a native German speaker, marked every relevant article on these pages before any system output existed and without seeing it. She drew a rectangle around each article in the PDF and added a short note. Two rules guide her. First, the topic decides, not the word: an article that argues for European cooperation is IDEA even if it never says "Europa", and gets the tag *implicit*. Second, the article is the unit: it gets one grade and one stance. The grade is IDEA as soon as any part of the article discusses the idea. MENTION marks an article that names Europe, Paneuropa or a related name in a recognisable way but is about something else. Stance, for IDEA only, is PRO, CONTRA, NEUTRAL or UNSURE. A note *none* marks a page that was read and holds nothing, so read pages can be told from unread ones. A passage belongs to an article if at least half of its area lies inside the article's rectangle. The full guideline is in our repository (`docs/annotation_guideline.md`).

One annotator. One historian marked all pages. We planned a second annotator on 8 of the 20 tier-1 pages, but a second historian could not be found in the time available. The reliability of the labels is therefore not measured, and relevance judgements differ between people even in large test collections [31]. We keep all claims modest and return to this in Section 7.

Measures for RQ1. Each method ranks all passages of one tier at a time, the pages a historian has in hand: the 608 passages of tier 1, and separately the 288 of tier 3. As a baseline without any tool, R0 keeps the reading order: the pages read from top to bottom. An article counts as found as soon as any one of its passages is read, and only IDEA articles count as relevant. Our main measure is *reading effort*: the number of passages a historian reads, from the top, until she has found 80% of the IDEA articles. It follows the idea of Cooper's expected search length [4]: what matters to the user is how much she must read. We also report article recall at 5, 10 and 20 passages read, and recall on the *implicit* articles, which keyword methods cannot find by construction. As a secondary, passage-level measure we report nDCG@10 [15]. It rewards a ranking that puts the most relevant passages in its first ten places, and gives less credit the lower they appear. An IDEA passage has gain 2, because

[2]https://docs.hpc.gwdg.de/services/ai-services/chat-ai/models/index.html and https://docs.hpc.gwdg.de/services/ai-services/chat-ai/faq/index.html, checked 30 Sep 2026.

it is what the historian looks for; a MENTION passage has gain 1, because it names the idea and may lead to the debate, but is not the debate itself. Where the historian marked the IDEA part of a long article, only that part has gain 2 and the rest of the article has gain 1. We also break the results down by OCR quality, to see whether noisy text hurts some methods more than others; the measure was fixed before we ran it. A passage's quality is the share of its words that a German word list knows (`wordfreq` 3.1.1; words of three or more letters, no numbers). Historical spellings such as *Theil* pass, while garbled words such as *bölferbund* fail. An article's quality is the mean over its passages, weighted by words. We split the tier-1 IDEA articles at the median quality and compare, per method, where each half is first found.

Measures for RQ2. We compare the model's stance with the historian's on her 16 IDEA articles in a confusion table and with Cohen's κ [3]. κ corrects the share of agreement for the agreement expected by chance, which matters because most articles are NEUTRAL. Articles she marked UNSURE are reported apart. The quote check compares each quote with the article text after folding case and spaces. A quote is *exact* if it occurs as it stands, *close* if a fuzzy partial match reaches 90 of 100, and *not found* otherwise. For example, a quote in which the model writes *Interesse* where the OCR reads *iyereffe* is no longer exact. With 16 articles, 15 of them with a decided stance, we report counts and keep the claims modest.

Controls against bias. Every tier-1 extract was returned by an Arcanum search for "paneuropa" and contains a form of the word. Comparing methods by whole extracts would therefore tell us nothing: the keyword would always be right by construction. We therefore score passages within pages, and tier 3 shows how the methods behave on pages nobody chose. The word list, concept descriptions and settings were fixed and committed before any label came back. One extra row, B0-literal, was added after the first results; we mark it as not pre-registered. There is no separate test split, so we tuned nothing on the results.

6 Results

Gold standard. The historian read every page of tiers 1 and 3. In tier 1 she found 15 IDEA and 6 MENTION articles; 2 of the 20 pages hold neither. Of the 15 IDEA articles, 9 are NEUTRAL, 4 PRO, 1 CONTRA and 1 UNSURE. In tier 3, 9 of the 10 pages hold no relevant article; the tenth holds one IDEA article, on a European currency.

RQ1: ranking. Table 4 gives the results of all methods on both tiers, and Figure 5 the full reading-effort curves on tier 1. On the keyword pages of tier 1, reading in page order needs 444 of 608 passages to reach 80% of the 15 IDEA articles. Every ranking method cuts this to between 35 and 62 passages. The archive keyword B0 needs the least reading, 35 passages, and it also finds the most articles in its top 20. The wider word list and OCR tolerance do not help here: B1 needs 41 passages and S1 58. Dense retrieval S2 needs 60 passages, but of the pre-registered methods it gives the cleanest top ten (nDCG@10 0.93) and the second shortest route to all 15 articles (101 passages, after B0 with 90). Fusion S3 adds nothing over its two parts.

B0 matches the stem *paneurop**, as the archive search did. Half of the keyword extracts, 7 of 14, contain only inflected forms such as *paneuropäischen*, yet the archive returned them. After the first results we added a literal keyword, *paneuropa** (B0-literal), to check whether the stem flatters B0; this row is not pre-registered. The literal keyword is brittle against German word endings. It needs 141 passages to reach 80% of the IDEA articles and 451 to reach all of them, against 35 and 90 for the stem. Four congress reports that use only the adjective fall far down, the last to passage 451. Its top ten is still the cleanest (nDCG@10 0.97): what it matches is relevant, but it matches too little.

Table 4: RQ1. Effort: passages read to find 80% (E80) or all (E100) of the IDEA articles; lower is better. R@k: share of IDEA articles found in the top k passages. Impl.: *implicit* articles found in the top 20, of 3. Tier 3 has one IDEA article; Rank is its first position. Bold: best pre-registered method. †Added after the first results; not pre-registered. Source: `runs/rq1_results.json`.

	Tier 1 (keyword pages)						Tier 3
Method	E80	E100	R@10	R@20	Impl.	nDCG	Rank
R0 order	444	604	.07	.07	1	.65	238
B0 keyword	**35**	**90**	**.40**	**.60**	0	.88	238
B0-literal†	141	451	.40	.47	0	.97	238
B1 exact	41	178	.27	.53	1	.72	**1**
S1 fuzzy	58	125	.20	.47	0	.80	2
S2 dense	60	101	.20	.47	0	**.93**	4
S3 fusion	62	134	.20	.40	0	.92	**1**

The three *implicit* articles are a negative result. All three sit on one page, and no ranking method puts more than one of them in its top 20. Dense retrieval, the method meant for them, finds the first at passage 60. Only B1 has one in its top 20, and so has reading order, because that page comes early.

On the control pages of tier 3, the picture turns. B0 matches no passage at all, so the one IDEA article, on a European currency, stays at passage 238, as in reading order. B1 and S3 rank it first, S1 second and S2 fourth. This rests on a single article.

The keyword pages are also much noisier than the control pages. In the median passage, 0.68 of the words are known German words on tier 1, against 0.90 on tier 3. We split the 15 tier-1 IDEA articles at the median quality, 7 low and 8 high. Figure 4 shows where each article first appears in each ranking. The keyword and the word lists find the cleaner half earlier. Under B0, a high-quality article is first found at passage 8.5 in the median and a low-quality one at 24; its top 20 holds 6 of 8 against 3 of 7. Dense retrieval shows almost no link to OCR quality (Spearman's $\rho = -0.09$). On the noisy half it is the best method, with a median of 16 and 4 of 7 articles in its top 20. The OCR-tolerant list S1 does worst on the noisy half, with a median of 46. With 15 articles, these numbers describe; they do not test.

RQ2: stance with evidence. The model answered all 16 IDEA articles of tiers 1 and 3 in the required format. Without the one article the historian marked UNSURE, it agrees with her on 7 of 15 (Cohen's $\kappa = 0.13$, Table 5). It recognises all 5 PRO articles, but it reads 7 of the 9 NEUTRAL articles as PRO. The historian then reviewed every disagreement together with the model's reason. Two patterns

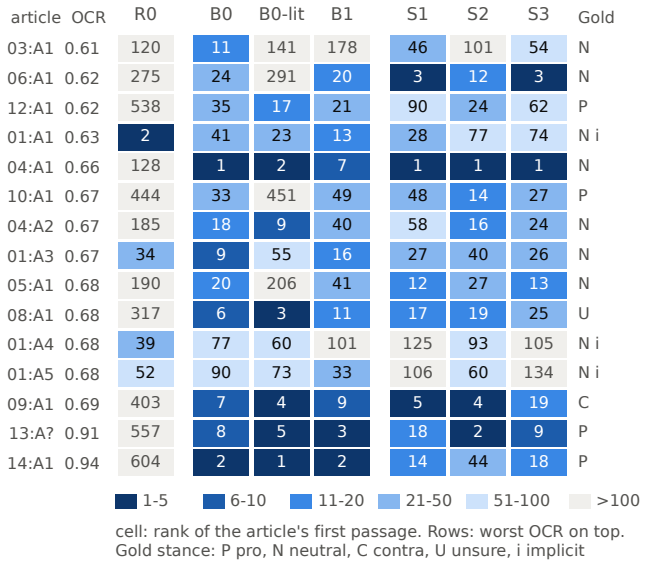


Figure 4: Where each tier-1 IDEA article first appears in each ranking (rank of its first passage; darker = earlier). Rows (page:article, without the "T1-" prefix) are sorted by OCR quality, worst on top. Right: the historian's stance (P pro, N neutral, C contra, U unsure; i = *implicit*).

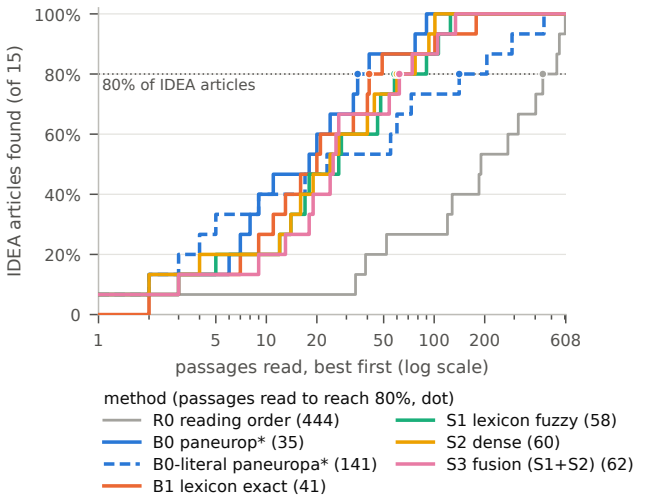


Figure 5: Reading effort on tier 1 (608 passages, 15 IDEA articles): share of articles found against passages read, best first (log scale). Legend numbers: passages read to reach 80%. R0 is reading the pages in order.

explain the 7 cases. Three are reports of the 1926 congress that summarise pro-European speeches. The model takes on the speakers' voice, although the prompt states that the text's own voice counts. The other four deal with French–German reconciliation, and the model reads it as support for European unity. For the historian, a rapprochement between two old enemies is not yet the European

Table 5: RQ2. The model's stance against the historian's on the 16 IDEA articles of tiers 1 and 3. Agreement on the 15 decided articles: 7 ($\kappa = 0.13$). Quote check: 5 exact, 7 close, 4 not found. Source: `runs/rq2_results.json`.

Historian ↓ / model →	PRO	CONTRA	NEUTRAL
PRO (5)	**5**	0	0
CONTRA (1)	0	**0**	1
NEUTRAL (9)	7	0	**2**
UNSURE (1)	0	1	0

idea. She labels such an article PRO only when the text itself ties it to Europe, as one article does by calling it the "cornerstone of European reconciliation". The one CONTRA article, a dismissive review of the congress, becomes NEUTRAL. The UNSURE article rejects a French-led Paneuropa but argues for a German-led Central Europe; the model calls it CONTRA, from a quote taken out of context. In her review, the historian finds the model's label defensible in 3 of the 8 disagreements (two French–German cases and the dismissive congress review), not defensible in 4, and leaves one open. Counting the defensible ones, 10 of 15 labels would be acceptable to her; since she judged them after seeing the model's answer, this is not an independent measure. We also checked whether each quote really stands in the article. Of the 16 quotes, 5 appear word for word, 7 with small differences, and 4 not as they stand. These four are not copied exactly. The model joined pieces of text with "...", or wrote a clean word where the OCR has a garbled one, such as *Interesse* for *iyereffe*. Each piece still matches a stretch of the article closely (character similarity 0.77 to 0.99), so none appears invented. The lowest match comes from a passage whose OCR lost its spaces.

Examples. Two cases show the typical behaviour behind these numbers: one for the stance and one for the ranking. A report of the 1926 congress (T1-05) summarises a speech on a "paneuropean economic union" of twenty states without tariffs. The model quotes the programme and calls the article PRO; the historian reads a neutral report of what the speakers wanted. This is the speech-voice error. A congress report on page T1-06 (Figure 3) uses only the adjective, in garbled forms such as *europäifche*. The OCR-tolerant list matches these forms and ranks the article third, while the literal keyword reaches it only at passage 291. This is where OCR tolerance pays off.

7 Discussion and Reflection

What the results mean for historians. On pages that an archive search returned, ranking pays off: any method cuts the reading to 80% of the relevant articles by a factor of seven or more. But the plain keyword, used as a stem as the archive does, is hard to beat there. Arcanum is an archive whose OCR and search historians already rely on; on archives with weaker OCR or a literal search, the keyword would likely do worse. A literal keyword without the German endings loses much of that advantage. Most relevant articles on these pages report the first Paneuropa congress in Vienna (October 1926) and name it, which is the selection bias we expected. A historian who already holds keyword pages gains most from simply reading the keyword passages first. The picture changes on

pages nobody selected. There the keyword finds nothing, and the word lists and dense retrieval surface the one relevant article at once. This is the case that matters for finding what a keyword search never returned, but it rests on one article. Of the pre-registered methods, dense retrieval gives the cleanest top ten, yet it does not find the articles that avoid the word "Europa". So a historian should not trust any of these methods to find the implicit debate; with the current system, she still has to read. OCR noise hurts the keyword and the word lists more than dense retrieval, which may be because meaning survives a misspelled word. The OCR-tolerant list does not solve this; its fuzzy matches may also pull in unrelated garbled words. With 15 articles, both points are hints for the next study, not findings.

The stance reading is not good enough to replace the historian's, but in this context its errors have two clear sources. The model never misses a PRO article, yet it takes a reported speech for the paper's own voice. This is exactly the distinction that the guideline's stance rule draws. It also reads French–German reconciliation as support for European unity. Our guideline does not draw that line explicitly, so the next guideline and prompt should state it. The historian finds 3 of the 8 disagreements defensible, so part of the gap lies in the task definition, not only in the model. Historians may also disagree with each other on these lines; with one annotator, we could not see how much. So the model's stance can serve as a pointer to check, not as a verdict. Its reason shows why it decided. The quote check is cheap and flags every quote that is not verbatim, which matters because the historian must find the words on the page.

From a planned system to a tested slice. Our design dossier planned a four-stage system for the press of interwar Romania, 1918 to 1939. It would prepare the archive with OCR quality scores and language detection, and then find candidate passages. It would characterise them by topic, stance, actor and voice, and let historians explore the results down to the scanned page. Its checks included a rise of the debate around Briand's plan from 1929 to 1932, and a comparison of the Romanian-, German- and Hungarian-language press. What we built is one thin slice of it: the ranking step and the stance with evidence, on 30 pages of the German-language press.

Two reasons made the plan shrink. First, data: Arcanum does not allow mass downloads, and our subscription ended in June, so we worked with the 19 extracts and the 50-page run we had downloaded by hand. These were already a selection, not all the pages the search for "paneuropa" returns, and they cover a few months of 1926 and 1929, which rules out any timeline. Second, labels: no gold standard existed, so building one came first and bounded everything else. We kept what the dossier called its key idea, the funnel instead of a classifier, and its demand for reasons, not verdicts.

Working with the historian changed the study more than we planned. Her revision of the guideline introduced the rule that the topic decides, not the word, and made the article the unit. We then changed our scoring from passages to articles, because a historian counts articles. We also caught a bias in our own preparation. Our first test rectangles, drawn to try the code, sat on articles that contained "Paneuropa", which is exactly the keyword bias RQ1 measures. The first page packs we sent still carried these marks; we rebuilt them without any marks before the historian started. A dry run before any labels existed revealed a flaw in our word list: the phrase "Vereinigte Staaten von Europa", split into words, also matched the United States. We removed it before the queries were fixed.

Limitations. Our pages were mostly selected by the keyword we compare against, so tier 3 is the only unbiased sample, and it is small. The gold standard is small, and it has one annotator and no measure of agreement. We planned a second historian on 8 pages, but none could be found in the time available. So we do not know how far another expert would draw the same articles or give the same stances, and every result here rests on one person's reading. We wrote the word list and the concept descriptions ourselves; historians should define what counts as the idea, and a list they review could change the results. We study three German-language titles of interwar Romania only; our results say nothing yet about the Romanian- or Hungarian-language press. Our 1926 pages come mostly from a Saxon paper in Hermannstadt and our 1929 pages from a Swabian paper in Temeswar, so differences between the years cannot be separated from differences between papers and regions. We kept the Fraktur OCR noise rather than correcting it; this is realistic, but a different OCR would change the numbers. The stance results rest on one open-weight model and one prompt, run once on GWDG's SAIA service, and on 15 articles with one annotator's labels. Reading effort counts passages, not minutes; we ran no user study.

Next steps. A study with historians would show whether fewer passages really mean less time and more trust. The same funnel can run on the Romanian and Hungarian press, each with its own word list and a small gold set. With a longer run of pages, the rise of the debate after 1929 becomes testable against the historical literature. A second historian should mark a sample of the pages; κ per passage, with and without *borderline* articles, and stance agreement on the articles both mark as IDEA would then measure the labels. That stance agreement would also give a human ceiling for the model's κ of 0.13. Reprint detection and an OCR quality flag next to each result are the next features the historians asked for.

8 Conclusion

We asked how far ranking can shorten a historian's reading after an archive keyword search. We also asked how well an open-weight model reads the stance of what she finds. On keyword-selected pages, every ranking method shortens the reading by a factor of seven or more, and the archive keyword itself shortens it most. On pages nobody selected, only the methods beyond the keyword find the relevant article, and they rank it near the top. No method finds the articles that discuss the idea without naming Europe, so these still need a human reader. The model agrees with the historian on only 7 of 15 stances. It recognises support for the idea, but it mistakes reported speeches for the paper's own voice and French–German reconciliation for the European idea, and 4 of its 16 quotes are not verbatim. For tools for historians, this suggests a division of labour: ranking saves reading, and a model may propose a stance only together with a quote she can check.

Contributions

Bea Dippold revised the annotation guideline (the rules that the topic decides and that the article is the unit), marked the gold standard, wrote the history background and checked the model's stance reasons. **Simon Manzenberger** documented the historians' requirements, designed the system architecture and the historian's workflow, and reviewed the paper. **Ruo Shan Tan** led the project and the repository, made the scoping decisions, ran the stance model and wrote the paper.

Repository and Resources

Code, configuration and documentation: https://github.com/tanruoshan/popline. We created the annotation guideline (v3) and the pre-registered queries. We list every page used with its Arcanum file name (title, year, month and archive identifier) and page range, so the page sets can be rebuilt with archive access. The repository also holds the passage labels derived from the annotations, and the code for passages, ranking, stance and evaluation. The newspaper pages themselves are subscription material and are not in the repository. With the pages in place, one script (`run_all.ps1`) reruns every step from the PDFs to the tables and figures, and each step records the code version, settings and inputs it used; 27 tests, one of them end to end, check the parts.

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A Stance prompt

The model receives one system message and one user message per article (`src/popline/stance.py`). It is not offered UNSURE. An answer in the wrong format counts as a failure and is never retried or fixed by hand; only network errors are retried. All 16 answers were in the right format.

System. *You are a careful historian's assistant. You read German-language newspaper articles from interwar Romania (1920s) and answer only in the JSON format the user asks for.*

User. *Below is one newspaper article. The text comes from OCR of an old Fraktur print, so it contains recognition errors; read past them. What is the stance of this article toward the idea of European unity or cooperation? Stance toward the idea of European unity or cooperation itself (not toward a person, a country or a conference): PRO: the text supports or welcomes it, or presents it as desirable or necessary; CONTRA: the text rejects, mocks or warns against it; NEUTRAL: a report without a judgement, or balanced pro and contra. The voice that counts is the text's, not that of the people it quotes: a neutral report of a pro-Europe speech is still NEUTRAL. Answer with one JSON object and nothing else:* `{"stance": "PRO" | "CONTRA" | "NEUTRAL", "reason": "<one English sentence>", "quote": "<the passage your answer rests on, copied exactly from the article as printed below, at most 40 words>"}` *Article:* followed by the article text.